INTERMEDIATE REPORT

1. General Information

DFG reference number: AOBJ: 681851

Project number: EL 918/2-1

Project title: **Welt-Wärmestrom-Datenbank-Projekt**

World Heat Flow Database project

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Reporting period: 01.04.2022 - 31.08.2024

Entire funding period: 14.04.2022 - 13.04.2025

2. Summary

The World Heat Flow Database (WHDB) project aims to develop a modern research data infrastructure (RDI) for global terrestrial heat-flow data, which are essential for understanding Earth's thermal field, geodynamic processes, and geoenergy resources. The new heat flow RDI enables extended interpretations using contemporary data analysis and visualization tools, providing the community with efficient access to a century of global research and data. This intermediate report summarizes activities and developments from the project start to month 27 of the 3-year project (2022–2025). Key achievements include: (1) developing a new RDI that serves as a standalone reference for data, related publications, projects, and researchers, (2) reassessing, enriching, and updating the Global Heat Flow Database (GHFDB) to enhance scientific standardization and usability, making it ready for data analytics and digital research methods. The RDI (3) adheres to FAIR and OPEN data policies, is technically interoperable with other data services, and is integrated into NFDI4Earth activities with connections to globally used persistent identifiers like DOI, ORCID, IGSN, and ROR. Data quality is ensured through standardization and reviewing hundreds of publications to enrich metadata. As of month 27, 52% of the GHFDB is quality-controlled, with 60% expected by project end. The prototype launch is anticipated in Q1/2025.

Zusammenfassung

Das Welt-Wärmestrom-Datenbank-Projekt (WHDB) entwickelt eine Forschungsdateninfrastruktur (FDI) für terrestrische Wärmestromdaten, die grundlegend für das Verständnis des Erdtemperaturfeldes, plattentektonischer und geodynamischer Prozesse sowie der geoenergetischen Nutzung sind. Die neue FDI ermöglicht erweiterte Interpretationen durch Datenanalyse- und Visualisierungstools und bietet effizienten Zugriff auf 100 Jahre globaler Forschung und Datenerfassung. Dieser Zwischenbericht fasst die Entwicklungen bis Monat 27 des dreijährigen Projekts (2022–2025) zusammen. Wesentliche Ergebnisse sind (1) die Entwicklung einer neuen FDI als zentrale Anlaufstelle für Wärmestromdaten und verwandte Veröffentlichungen, Projekte und Autoren und (2) die Überarbeitung, Erweiterung und Aktualisierung der Datenbank zur Verbesserung der wissenschaftlichen Standardisierung und Nutzbarkeit globaler Wärmestromdaten. Die FDI (3) folgt FAIR- und OPEN-Daten-Prinzipien, ist kompatibel mit anderen Datendiensten, in NFDI4Earth-Aktivitäten eingebettet und verknüpft mit persistenten Kennungen wie DOI, ORCID, IGSN, ROR. Die Datenqualität wird durch (Metadaten-)Standardisierung und die Überprüfung hunderter Veröffentlichungen sichergestellt. Bis heute (M27) sind 52 % der globalen Wärmestromdaten qualitätsgeprüft (Ziel: 60 % in M36). Die Veröffentlichung des Prototyps des Datenportals ist für Q1/2025 geplant.

3. Progress Report

3.1 Background and objectives of the project

Background - heat flow

Heat-flow density data, representing the thermal energy flowing from the Earth's interior to its surface, provides essential insights into our planet's structure, tectonic and geodynamic processes, and geoenergy resources. This data is crucial for analyzing the Earth's temperature field in both continental and marine settings. Heat-flow values are systematically reported for sites investigated during industrial exploration (oil, gas, geothermal) or scientific drilling projects like IODP or ICDP. These data are prerequisites for the thermal modeling of geothermal systems, hydrocarbon maturation analyses, and safe geological storage of energy and waste. Heat-flow data are critical for preventing and managing potential subsurface utilization conflicts affecting the thermal field. Without reliable heat-flow data, many scientific thermal analyses and industrial applications lack a critical input parameter. Heat flow cannot be measured directly; it is a derivative measure calculated on determined temperature gradients and thermal conductivities, calling for comprehensible documentation of involved data and metadata for its evaluation.

Background - challenge

The status of data and documentation in past global compilations did not reflect the parameter's importance. Data structures adhered to outdated concepts from 1975 (cf. Jessop et al., 1976). Data were often extracted directly from research articles, documenting only basic information on heat flow and coordinates. Consequently, documentation of heat-flow (meta)-data was inconsistent and incomplete, and data values frequently lacked source references (cf. Fuchs et al., 2022). Methodological differences in determining heat flow further complicated data integration and missing metadata were a general problem in the global database (Global Heat Flow Compilation Group, 2013). Existing German research data infrastructures (RDI), like 'GeotIS' and 'FIS Geophysics' neither contain or plan to incorporate heat-flow data (personal communication, Agemar, 2022) nor aim for global coverage. On a global scale, the International Heat Flow Commission (IHFC) was established in 1963 to maintain the Global Heat Flow Database (GHFDB) (a spreadsheet data collection). Techniques and knowledge for determining heat flow have evolved, affecting calculation methods and metadata reporting. Consequently, the IHFC did not provide an authenticated, quality-controlled, modern database or RDI and never attempted to review the existing data collected for several decades thoroughly. This situation hampered scientific research on the Earth's heat flow and sparked debate for years (e.g., Artemieva & Mooney, 2001; Hasterok & Chapman, 2011; Lucazeau, 2019). A systematic collaborative approach at the global scale was needed to provide a RDI featuring (1) quality-controlled heat-flow data, (2) a consistent storage concept, (3) user-friendly mapping and visualization interface (4) comprehensible documentation of data and metadata, and (5) a quality evaluation scheme.

Objectives

With the *World Heat Flow Database* Project, we proposed developing this new heat-flow RDI that offers a modern data portal with quality-assessed, up-to-date, well-documented, extended, and restructured heat-flow data. The data portal is designed as a reference for heat-flow data and information, including authors, researchers, publications, and projects.

The **main objectives** for the first project phase (36 months, 04/2022 – 03/2025) are:

1. to develop a research data infrastructure that offers a standalone reference and one-stop shop for heat-flow data, and related publications, projects, and researchers.
2. to provide a **reassessed**, enriched, and updated Global Heat Flow Database **by reviewing the global literature and** improving the scientific **standardization** and usability of global heat-flow data, making it ready for data analytics and digital research methods.
3. to establish a **user-oriented research data and information platform** that reflects FAIR and OPEN data policy that is technically **interoperable** with other geoscientific data services **embedded** into the NFDI4Earth activities, and is interconnected to globally used persistent identifiers, like DOI, ORCID, IGSN, etc.
4. to provide an easy-to-use, but fundamental **scientific service** of quality-proofed and curated heat-flow data to scientists, engineers, policymakers, and the general public.

3.2 Work Steps

Data management and collaborative enrichment (lead: GFZ 4.8)

The core aspects of WP1 were (1) strengthening the heat-flow community and (2) collaboratively assessing the GHFDB. To analyze community expectations and requirements (Task 1.2), we organized workshops, online formats, scientific meetings, stakeholder forums, and feedback schemes to engage with the global heat-flow community (Task 1.4; cf. section 3.5). These activities identified key needs in data and metadata quality, tools, services, knowledge transfer, and capacity building (Task 1.1). To ensure reported heat-flow data was accompanied by sufficient metadata for quality assessment, we developed new community-driven protocols and metadata schema (Task 1.5; D1.1) and implemented data quality measures (Fuchs et al., 2023; D1.2). New data **upload templates** were created to report heat-flow data according to the schema (IHFC et al., 2024), which comply with FAIR principles and include persistent identifiers (PIDs) like ORCID, ROR, IGSN, and DOIs. The schema also utilizes published controlled vocabularies (SKOS/RDF) to support metadata interoperability and includes properties for data quality assessment. Additionally, new codes for automatic vocabulary checks and quality scoring were developed and published (Chishti et al., in prep.).

A new global data set was initially compiled from existing sources (IHFC, 2012; Lucazeau, 2019; ThermoGlobe, 2021) and approximately 500 new publications (Task 1.6). All primary references (about 1,600) were reviewed to meet the new database schema, adding quality flags and additional variables. The 2024 **database release** (Global Heat Flow Data Assessment Group et al., 2024; D1.3; D1.4) contained assessed and improved data from 800 publications, filling 85% of the metadata fields, up from 15% in the 2021 version. This effort, led by the project team and supported as part of the *Global Heat Flow Data Assessment Project* (website), was guided by IHFC and Task Force 8 of the International Lithosphere Program (ILP). Additional external funding for two Postdoc positions was secured to manage this process. Further information on references, authors, and researchers is stored in interlinked tables (Task 1.6), with frequent database releases at GFZ Data Services (in 2021, 2023, and 2024). The general assessment (Task 1.7) includes quality controls, like reviewing all primary sources according to the new metadata schema and ensuring data extraction follows the "four-eyes principle." New submissions to the portal undergo multi-level validation, including database, application, and manual checks. Ongoing efforts focus on updating extending the database (Task 1.7) to achieve a 60% update of the initial global dataset by M36 (D1.5), with 52% already completed seven months before the project ends.

Database, portal back end, architecture and interoperability (lead: GFZ LIS)

WP2 involved designing and conceptualizing the research data infrastructure, focusing on (1) front-end and back-end interfaces, (2) their interaction strategy, and (3) the workflow for the new data submission system. The architecture, libraries, and communication protocols are outlined in Fig. 1. A core data model was developed (Task 2.2) to integrate the portal database with established metadata standards (RAiD, DataCite, IGSN) and FAIR principles, including persistent identifiers (PIDs). The new database (D2.1) connects each heat-flow measurement to its original literature reference (via DOI), authors (via ORCID), and related physical samples (via IGSNs). A basic API (Task 2.4) was built for metadata and data harvesting by external parties. Integration with GFZ Data Services supports data publication and archiving, linking the portal with workflows for research data interoperability (Task 2.5).

The web portal (D2.2, D2.3, D2.4) for the GHFDB, developed (Task 2.6) using Geoluminate (https://github.com/Geoluminate/geoluminate) offers several key features:

1) A consistent data model that promotes integration with existing initiatives including DOI, DataCite, ROR and ORCID, thereby maximizing findability, accessibility, interoperability, and reusability under the **FAIR** data principles.

2) A streamlined user authentication system, account management, access control and permission systems that maintain portal security, user privacy and data integrity.
3) A fully-featured administration site that allows trusted users to manage the portal and review new submissions.
4) A set of predefined configuration files that facilitate deployment of the research portal to remote or self-hosted production environments.

The codebase for the GHFDB web portal (https://github.com/ihfc-iugg/ghfdb-portal) focuses on defining heat flow-specific data models, as per the community-defined recommendations of Fuchs et al., (2023), and includes tools for deploying the portal using Docker and Kubernetes technologies ensuring compatibility with various infrastructures. The inclusion of configuration files in the portal repository allows it to be deployed on various IT infrastructures: from PC to cloud-based services like Google Cloud and Amazon Web Services. This flexibility ensures that data and metadata standards from our project can be utilized by the international heat-flow community, regardless of legal or technical constraints that may prevent some community members from contributing directly to the GHFDB.

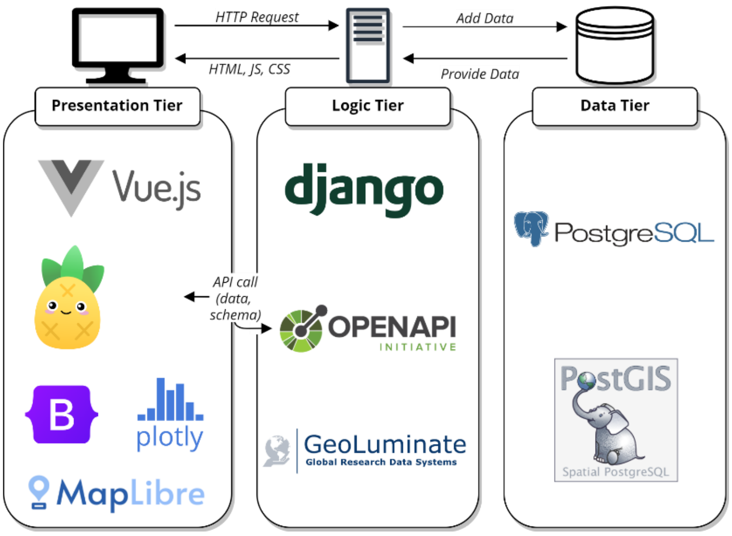

Figure 1: System architecture illustrating the libraries used and the communication between the tiers

The configuration files define several services that make up the backend architecture and are critical to the deployment and functionality of the web portal. In a production environment, these services include: (1) The web application; (2) PostgreSQL, a relational database for data and metadata storage, with PostGIS for geospatial data manipulation; (3) Redis, an in-memory data store used as a cache and message broker for background tasks; (4) Celery, a distributed task queue for asynchronous tasks such as sending emails and processing files; (5) MinIO, an Amazon S3-compatible object storage server for user-uploaded files; and; (6) Traefik – a reverse proxy and load balancer that routes incoming requests to the appropriate service.

The official instance of the GHFDB web portal is hosted on internal servers maintained by GFZ. Media files are stored alongside in a single-server setup as private and secure by default preventing unauthorized access. Automated **backups** of the database and media files occur at regular intervals, and the GHFDB collection (as a separate product) will be **periodi-**

cally released (in different formats; Task 2.7), transferred to GFZ Data Services and formally published and archived with DOI (cf. WP1). *At the time of reporting, the submission system (D2.2) interoperable with GFZ Data Services, that allows the semi-automatic generation of metadata for citable data publications are still under construction. Likewise, work on the review area (internal revision system; D2.3) is open. The prototype release (D2.6) is scheduled for M35.*

Front end, data exploration, visualization and mapping (lead: TU Dresden)

In WP3, the main achievement is the development of a web platform that integrates several key components: (1) a project webpage, (2) a web-based data portal with user account management, (3) a community area, (4) a data submission system, (5) an interconnected literature and author database, (6) a graphical user interface, and (7) a web-based map viewer for exploring heat-flow data. The Web portal has been developed using the Django Web Framework. The portal (along with Docker configuration files), built with the Django Web Framework, is open-source and available on GitHub (https://github.com/ihfc-iugg/ghfdb-portal). The map viewer, based on a Vue.js front-end, offers data selection, search functions, statistical visualizations, and analytical tools. Users can search based on categories, keywords, and spatial-temporal extent, with options for customizable map regions and data classification using algorithms like quantile and Jenks. The data submission system supports data uploads either manually through a form or automatically via file import. Uploaded data undergo technical validation for consistency, completeness, and accuracy, forming the basis for DOI publication with GFZ Data Services. (D2.2). Throughout the platform, we established a feedback-and-commenting system (D2.4) that allows users to add commentaries and annotations to data or projects on our platform, which will help to collect observations and experiences from researchers worldwide. *Still under construction* are *(1) the FAQ and help section as central support information or helpdesk with structured contents (for instance as a decision tree) to interactively assist the users in all matters or relevance to the heat-flow database.*

Dissemination, outreach, and project management (lead: GFZ)

WP 4 ensured (1) coordination, execution, and follow-up of the scientific program, (2) the management of time and financial resources allocated to the project, (3) the coordination and implementation of the collaborative work of the international community, (4) communication, dissemination, outreach, and (5) adequate exploitation of the project results to the scientific community, the public and the stakeholders. Dissemination and outreach results are listed in Sections 3.5 and 4. All scheduled activities were conducted without significant deviation.

3.3 Availability of Results and Reuse Options Compatibility

All data are openly provided as regular data releases in a domain repository for geosciences **without retention** periods (usually released once a year during project duration; e.g. Global Heat Flow Data Assessment Group, 2023; 2024). The heat-flow database with 91k data points (Release 2024) is mainly **fed from existing studies** (1939–2024) and compiles data from ca. 1,568 different studies. We retrieve data only from publications and always link the respective publication(s) with the values in the database. Public releases of the GHFDB are published with DOI and archived in the domain repository GFZ Data Services under a CC BY 4.0 license. No restrictions are foreseen or expected regarding the future usability of these data. The basic professional protocol and metadata schema (Fuchs et al., 2023) was published OA and might be developed further in the future. Our **data are already used** by third-party applications (e.g. by GeoMap App, by GeoMap of the Project InnerSpace), Already before prototype release, the codebase is published and can potentially be reused.

3.4 Compatibility with national and international Infrastructures

One of the primary targets of our project is to establish a user-oriented research data and information platform that adheres FAIR and OPEN data policies, and ensures the provision of quality-controlled and validated heat-flow data. We offer freely accessible annual data releases with documentation of the changes that guarantee transparency, quality assurance, and version control. Methods suitable for implementation in the project were documented and discussed (Ott et al., 2024 is a review article on mapping techniques; Chishti et al.: Documentation of code in a GitHub repository). With well-defined metadata standards, our API allows data exchange, combability, and interoperability with other platforms. On a national level, conceptual exchange was achieved with **NFDI4Earth** (in the geophysics interest group; see also interactions at events like the EGU booth). We are closely interconnected with **GFZ Data Services** (project partner), and we provide released data to **GeotIS** (FIS Geophysics) and **PANGAEA** for implementation in their data and repository systems. With the recent and next release, we also strive to enhance data sharing by distributing our data towards larger international geoscientific data infrastructures, like **OneGeology, EPOS,** or **GEOSS**.

3.5 Project Outreach

Throughout the project, we (co-)organized several public or online scientific events for public outreach or scientific dissemination. For scientific dissemination, cf. section 4.

Events

Apr-Dec 2022: Online workshop series (4 meetings) on the development of the new metadata scheme, 45 participants. | **May 2022:** Public project kick-off meeting at GFZ (2 days) | **Jun 2022**: Cermak7 Meeting (7th International Meeting on Heat Flow and the Thermal Structure of the Lithosphere, 20-22 June, Potsdam; 68 participants) and writing workshop **| Jul 2022, 03–07**: 5-day IUGG summer school on heat-flow determination and data interpretation (at GFZ, 22 participants) **|Jul 2023,**

08–09: 2-day workshop on heat flow in Antarctica (at GFZ, 24 participants) | **Jul 2023, 10–11**: 2-day workshop on heat flow in Africa, South- and Central America (at GFZ, 23 participants) | **Sep 2023**: Online workshop series (3 meetings) on the development of the metadata scheme and white paper on heat flow as an essential ocean variable (EOV), 11 participants. | **Mar 2024**: Community and stake-holder kick-off meeting (online), 16 participants. | **Apr 2024**: EGU heat flow spring 2024 with stake-holder meeting as splinter meeting at EGU General Assembly and co-organization of joint meeting of Task Forces of the International Lithosphere Program (ILP) on the heat-flow assessment | **Apr 2024, 19-21:** Meeting and proposal writing workshop for the implementation of heat flow as an essential ocean variable (EOV) within the GOOS of the UNESCO

Science communication

Conference presentations (IUGG, GeoBerlin, EGU, AGU, etc.) | IUGG **newsletter** reports (July 2023, April 2024, July 2024) - reporting about summer school, workshops, EOV and project progress | **Website** communication (GFZ, IHFC, project, LI) | 24hrs **Global Heat Flow Day** 2022, 2023, and 2024 - One-day global online session with knowledge transfer and community-building aspects | YouTube **videos** on the data and meta data structure, the quality scheme (Feb 2024) or online work-shops, like the marine probe sensing workshop (July 2022). | **NFDI4Earth** (participation in interest group meetings since 2023, Live presentation at booth at EGU General Assembly, April 2024) | **Co-operation agreement** with the International Geothermal Association (IGA) on knowledge transfer and young researcher education | **Implementation** of our data into the geomapapp.org of Lamont-Doherty Earth Observatory (LDEO) of Columbia University, USA.; into the GeoMap of Project InnerSpace, USA; into the Geothermal Atlas for Europe (EuroGeoSurveys); and into the Quantarctica a free GIS for Antarctica, Geolocical Survey of Norway.

The results of our activities were reflected by **several regional to global media**:

scinexx.de ("Deutschlands Untergrund ist wärmer als gedacht", 02.12.2022) | Focus („Deutsch-lands Untergrund ist wärmer als gedacht", 03.12.2022) | Tagesspiegel ("Joker für die Energiewende: Der Wärmefluss aus dem Erdinneren ist offenbar größer als gedacht", 16.12.2022) | Umweltdialog.de („Deutschland: Mehr Wärme im Untergrund als bisher angenommen", 24.01.2023) | world-energy.org („Project InnerSpace Launches Collaboration for Global Heat-Flow Mapping ", Feb 2023) | thinkge-oenergy.com ("The IHFC has published the 2024 version of the GHFDB", 19.04.2024; "Project In-nerSpace launches collaboration for global heat-flow mapping" Feb, 2023) | Eavor.com ("IHFC fea-tures quality score evaluations for heat-flow data in 2024 report", 28.05.2024) | Linkedin.com – several posts and features since 2022

4. Publicly Accessible Project Results

Project website: www.heatflow.world/projects (online since 04/2022)

4.1 Publications with Scientific Quality Assurance

Peer-reviewed journals

Neumann et al. (in review): The 2024 status of the Assessment of the Global Heat Flow Database: Data, Tools and Publications. **Earth System Science Data** (OPEN ACCESS)

Fuchs, S., Norden, B., Neumann, F., et al. (2023): Quality-assurance of heat-flow data: The new structure and evaluation scheme of the IHFC Global Heat Flow Database, **Tectonophysics**, https://doi.org/10.1016/j.tecto.2023.229976 (OPEN ACCESS)

Fuchs, S., Förster, A., Norden, B. (2022) Evaluation of the terrestrial heat flow in Germany: A case study for the reassessment of global continental heat-flow data. **Earth-Science Reviews**, Volume 235, December 2022, 104231, https://doi.org/10.1016/j.earscirev.2022.104231 (OPEN ACCESS)

Peer-reviewed contributions to conferences

Neumann, F., Fuchs, S., Norden, B., Pazvantoglu, E., Elbarbary, S., Petrunin, A.G., Elger, K., Jennings, S., Frenzel, S. (in preparation) A new digital infrastructure for the Global Heat Flow Community. **Earth System Science Data**. (OPEN ACCESS) - submission in 09/2024

Ott, N.; Mäs, S.; Fuchs, S; Elger, K.; Jennings, S; Neumann, F.; Norden, B; Frenzel, S. (2024) Das Weltwärmestrom Datenbank Projekt: webbasierte Explorationswerkzeuge für Punktdaten, basierend auf VueJS und MapLibre. **FOSSGIS-Konferenz** (20. bis 23. März 2024, Hamburg, Germany), (ORAL) https://pretalx.com/fossgis2024/talk/CBWGLM/ (OPEN ACCESS)

https://doi.org/10.5194/egusphere-egu24-19849

Fuchs, S., Norden, B., Neumann, F., Global Heat Flow Data Assessment Group, International Heat Flow Commission (2023): The Global Heat Flow Database: New quality scheme and data release 2023, XXVIII **General Assembly of the International Union of Geodesy and Geophysics** (IUGG) (Berlin 2023). (ORAL) https://doi.org/10.57757/IUGG23-3313.

Jennings, S., Elger, K., Fuchs, S., Ott, N., Mäs, S., Norden, B., Neumann, F., & Frenzel, S. (2023). Heatflow.world: an online application for disseminating the Global Heat Flow Database to the international heatflow community. Deutsche Geologische Gesellschaft - Geologische Vereinigung e.V. (DGGV). https://doi.org/10.48380/KK40-NE65

Norden, B., Förster, A., Fuchs, S. (2023): The quality assessment of the German heat-flow database, XXVIII **General Assembly of the International Union of Geodesy and Geophysics** (IUGG) (Berlin 2023).(POSTER) https://doi.org/10.57757/IUGG23-3270

Fuchs, S., Förster, A, Norden, B. (2023) A new quality assessed heat-flow database for Germany, **DGG Jahrestagung 2022** Bremen, Germany, 06.03.-09.03.2023 (ORAL)

Neumann, F.; Norden, B.; Fuchs, S. (2022) The Global Heat Flow Data Assessment Project and Fellowship Program **RAUGM 2022**, Mexico, 30.10.-04.11.2022 (POSTER)

Fuchs, S., The Global Heat Flow Data Assessment Group (2022): The Global Heat Flow Database - status, progress and future projects, **Cermak7**: 7th International Meeting on Heat Flow and the Thermal Structure of the Lithosphere, Potsdam, Germany, 20-22.06.2022 (ORAL)

Norden, B., Förster, A., Fuchs, S. (2022) Quality assessment of German heat-flow data: a contribution to a revised Global Heat Flow Database, **Cermak7**: 7th International Meeting on Heat Flow and the Thermal Structure of the Lithosphere., Potsdam, Germany, 20-22.06.2022 (POSTER)

Norden, B., Fuchs, S., Neumann, F., Global Heat Flow Data Assessment Group (2023): The IHFC Global Heat Flow Database: New Structure and Evaluation Scheme for the quality-assurance of Heat-Flow Data - Abstracts, **AGU23,** San Francisco, USA (POSTER)

Other Publications and Published Results

Chishti et al. (in preparation): The heat flow quality code (OPEN ACCESS) – **GitHub**, submission in 09/2024

Petrunin et al. (in preparation): The Heat Flow Map of the Earth 2024, **GFZ Data Services** (OPEN ACCESS) - submission in 09/2024

Neumann, F.; Fuchs, S; Norden, B; Balkan-Pazvantoğlu, E; Petrunin, A; Elbarbary, S.; Jennings, S; Elger, K.; Frenzel, S.; Ott, N.; Mäs, S. (2024) Global Heat Flow Data Assessment Group (2024) The heat-flow data assessment project: Transformation of the Global Heat Flow Database. **EGU General Assembly 2024**, April 2024, Vienna, Austria (POSTER) https://doi.org/10.5194/egusphere-egu24-16952

Jennings, S; Elger, K.; Fuchs, S; Neumann, F.; Norden, B; Frenzel, S.; Mäs, S.; Ott, N. (2024) Geoluminate: A community-centric framework for the creation, deployment and ongoing development of decentralized geoscience data portals. **EGU General Assembly 2024**, April 2024, Vienna, Austria, (POSTER)

Global Heat Flow Data Assessment Group; et al. (2024): The Global Heat Flow Database: Release 2024. V. 1. **GFZ Data Services**. https://doi.org/10.5880/fidgeo.2024.014 (OPEN ACCESS)

Fuchs, S., et al. (2023): The Global Heat Flow Database: Update 2023. **GFZ Data Services** https://doi.org/10.5880/fidgeo.2023.008 (OPEN ACCESS)

Mäs, S.; Elger, K.; Fuchs, S.; Ott, N.; Jennings, S.; Norden, B.; Neumann, F.; Frenzel, S. (2023) Heatflow.world: an international research data infrastructure for the Global Heat Flow Database. **NFDI4Earth - Interessensgruppe Geologie & Geophysik (I3G) Workshop** 29.11.2023

Ott, N.; Mäs, S. (2023). Analysis of free web mapping libraries. **Zenodo**. https://doi.org/10.5281/zenodo.8139086 (OPEN ACCESS)

5. Further Information on the Project

In this project, we trained motivated young researchers who generated a number of research **data** (Global Heat Flow Data Assessment Group, 2023, 2024), **methods** (Ott and Mäs, 2023) **protocols, and metadata schema** (Fuchs et al., 2023), **software** (Geoluminate - at least support of it; codebase, GitHub, documentation), and **infrastructures** (codebase). Our researchers are open to new challenges and the project results are reusable and openly accessible to others.

5.1 Long-term Continuation and Sustainable Safeguarding

Even before this project was granted, the **GFZ committed to sustainably operate this data infrastructure** in-house, as attested by a letter signed by the CEO. This commitment includes providing all necessary financial (institutional funding), personal, and organizational support required for maintaining and continuously updating the infrastructure beyond the initial grant period. The IT and geoinformation department's extensive and diverse experience will fortify all safeguarding activities. Data curation will be continuously overseen by permanent staff (project PI). Frequent data releases will continue being stored in a widely recognized domain repository (GFZ Data Services).

5.2 Plans for the Continuation of the Work

Phase 1 will conclude with the release of the platform prototype and the database that is quality-controlled to a rate of about 60%. Phase 2 is crucial for the **consolidation of the technical database and the transition of the data portal** from prototype to deployment. It will also encompass to **complete the assessment** and metadata enrichment. Particular aspects of a follow-up study may involve (1) **harmonization and standardization** of oceanic heat-flow data and the development of metadata profiles for conductivity and temperature data (as supportive information) with the associated database extension; (2) **standardization** of data format and metadata profiles for grids from computational models; (3) completion of community-driven recompilation and metadata enrichment of globally unassessed heat-flow data and additional unpublished data (ca. 400 publications); and (4) standardization of global geological and surface data (e.g., crustal composition, elevation models, maps of sedimentation/erosion rates, mean surface temperature data, hydrothermal circulation,) to suitable grid scales for automated geographic cross-referencing with stored heat-flow data and associated corrections.

5.3 Project contributors

The quality check of the research data on an expert level is overseen by Florian Neumann (full-time Postdoc WP1), Sven Fuchs (PI), and Ben Norden (GFZ contribution). The technical data management at the database level and the backbone software development is conducted by Samuel Jennings (full-time Postdoc, WP2), with support from Simone Frenzel (GFZ contribution) and Kirsten Elger (PI). Upon project completion, data curation responsibilities will transition to Sven Fuchs, who is also the appointed database custodian of the International Heat Flow Commission (IHFC). This role is a subject of responsibility for his conversion to permanent employment intended to start in 2025/2026. Ben Norden, already in a permanent position within the same research unit, will continue to support data curation efforts. Kirsten Elger (permanent position) will continue to strengthen the links between the heat-flow data Portal and GFZ Data Services and provide metadata expertise.

German and international partners

A substantial number of national and international partner institutes participated in the World heat-flow database project, in particular in WP1, which focused on the collaborative enhancement and extension of research data. International institutions involved in the standardization process and the data assessment (2023; 2024) spanned across six continents:

Aarhus University, Aarhus, **Denmark;** CICESE, **Mexico;** Chevron Technical Center, **USA;** Chinese Academy of Science, Beijing, **China;** CONACyT, Morelia, **México;** CSIR, Hyderabad, **India;** Dokuz Eylül University, İzmir, **Turkey;** Dublin Institute for Advanced Studies, **Ireland;** Tebessi University, Tebessa, **Algeria;** ETHZ, Zurich, **Switzerland;** GEUS, Copenhagen, **Denmark;** Geological Survey of Japan, Tsukuba, **Japan;** Geological Survey of Israel, Jerusalem, **Israel;** Geological Survey of Slovenia, Ljubljana, **Slovenia;** IGME-CSIC, **Spain;** GEOMAR, **Germany;** GFZ Potsdam, **Germany;** GNS Science, **New Zealand;** Nanjing University, **China;** National Research Council, Turin, **Italy;** National Security Council, Putrajaya, **Malaysia;** Oregon State University, **USA;** Prince of Songkla University, **Thailand;** Ruhr University Bochum, **Germany;** Scripps Institution of Oceanography, San Diego, **USA;** Sorbonne University, CNRS, Paris, **France;** SMU Geothermal Laboratory, Dallas, TX, **USA;** The Ohio State University, **USA**; The University of Texas at Austin, **USA;** Universidad Nacional de Colombia, **Colombia;** Universidade de Evora, **Portugal;** University of Science and Technology, Krakow, **Poland;** University of Bremen, **Germany;** University of Genoa, **Italy;** University of Melbourne, **Australia;** University of Perugia, **Italy;** University of Tasmania, **Australia**; University of Tartu, **Estonia;** University of Helsinki, **Finland;** University of Genoa, **Italy**

Early career qualifications

During the project runtime, several academic qualification works have been supported.

Project staff

Samuel **Jennings** (M, GFZ Potsdam, 1st Postdoc position, Chief software developer), Florian **Neumann** (M, GFZ Potsdam, Postdoc, lecturing at University of Potsdam and during the project summer school, thesis supervision), Nikolas **Ott** (M, TU Dresden, 1st post-graduation position, software developer, teaching support and thesis supervision), Falko **Krügel** (M, TU Dresden, 2nd post-graduation position, development support) (Yui **Takahashi** (F ,TU Dresden, Master thesis), Holger **Ehrmann**, Ali **Mohammed** (M, GFZ Potsdam, Internships von FH Potsdam)

Early career participants (22) in summer school and workshops

Insaf Mraidi (F, **Tunesia**, Faculty of Sciences of Tunis), Magued Al-Aghbary (M, **Djibouti**, TU Freiberg), Song Zilong (M, **China**, Institute of Oceanology, Chinese Academy of Science), Xinzhe Zhao (F, **China**, QGESS Pty Ltd Australia), Elif Balkan-Pazvantoglu (F, **Turkey**, GFZ-Potsdam), Emma Chambers (F, England, Dublin Institute for Advanced Studies), Andrea Marchetti (M, **Italy**, University of Ferrara), Chrysanthi Pontikou (F, **Greece**, University of Pavia), Kim Lemke (F, **Germany**, University of **Lausanne**), Zanne Korevaar (F, **Netherlands**, Geologische Dienst Nederland), Maximilian Loewe (M, Germany, University of Edinburgh), Nishu Chopra (F, **India**, CSIR National Geophysical Research Institute, Hyderabad, India), Eswara Rao Sidagam (M, **India**, CSIR National Geophysical Research Institute, Hyderabad, India), Karina Fuentes-Bustillos (F, **Mexico**, Ensenada Center for Scientific Research and Higher Education), Harol Stiven Buitrago Segura (M, **Colombia**, Ensenada Center for Scientific Research and Higher Education), Ángela María Gómez García (F, **Colombia**, Geosciences Barcelona), Benjamin Norvell (M, **USA**, New Mexico Institute of Mining and Technology), Daniela Navarro-Perez (F, **Chile**, University of Leeds), Eskil Salis-Gross (M, **Sweden**, GFZ Potsdam), Carolin Wallmeier (F, **Germany**, GFZ Potsdam), Caroline Brand (F, **Germany**, University of Bremen), Judith Freienstein (F, **Germany**, University of Kiel)

Doctoral researchers involved

Several doctoral students supported our project with their doctoral projects. They received travel funding and joined our team as scientific guests. As part of their involvement, data publications and research papers have been published (cf. section 4).

Doctoral **researchers**	Doctoral **status**	**Start** of studies	**Funding** from WHDB project
Bustillos, Karina Fuentes	Ongoing; *CICESE, Mexico*	09/2022 – …	10/2022 – 01/2023
Mino, Belay Gulte	Ongoing; *University of Melbourne, Australia*	08/2023 – …	08/2023 – 10/2023
Rao, Sidagam Eswara	Ongoing; *CSIR, India*	01/2023 – …	07/2023 – 09/2023
Bencharef, Mohammed Hichem	Completed; *Tebessi University, Algeria*	06/2023 – ...	07/2023 – 09/2023
Buitrago, Harold	Ongoing *CICESE, Mexico*	05/2023 – …	05/2024 – 07/2024
Dutta, Ayan	Ongoing *CSIR, India*	08/2022 – …	03/2024 – 05/2024

5.4 Applicant's financial contributions

The GFZ supported the project group with the shared working time of two employees. This was mentioned in the proposal but was not allocated as an internal budget.

Norden, Ben Dr. 30% FTE, Scientist, Section Geoenergy – permanent contract,

Schöbel, Birgit 30% FTE, Scientist, Section Geoenergy – permanent contract,

The project PIs (Sven Fuchs, Kirsten Elger, Stephan Mäs) are running on institutional budgets (GFZ and TUD) and were substantially contributing to lead, management and conceptional developments of the project. Moreover, four GFZ scientists on third-party funding have supported the assessment work, namely Samah Elbarbary (100% FTE), Elif Pazvantoglu (100% FTE), Alexej Petrunin (25% FTE), and Julie Frieddel (25% FTE).

5.5 Community acceptance

Since the phase-one prototype release is scheduled for early 2025, detailed statistical data on user engagement, submissions, and growth metrics are not available, yet. However, following the recent questionnaire about our research data infrastructure in spring 2024 (March-April), we gathered valuable feedback from the community and stakeholders. We received feedback from 33 colleagues representing topic experts, library and information specialists, and geoinformatics. According to our survey, 90% of respondents expressed high satisfaction with the concept and the functionality of our data portal. Dr. Lesley Weyborn from the Australian National University stated, "This platform has substantially driven the capabilities by providing unprecedented access to high-quality heat-flow data". Building on this positive reception, our project has established a robust global collaborative network, forming partnerships with numerous geoscientific institutes that have significantly bolstered our data quality and review processes. Our data are already integrated in multiple prominent RDI (e.g., GeoMapApp, ArcMap Online, GeoMap, Quantarctica) and have been cited in over 50 peer-reviewed journals, including prestigious journals like "Earth-Science Reviews", or "Gondwana Research". These citations underscore the impact of our project on recent geoscientific research. During the project, we have hosted > 20 workshops that reached over 200 participants from the global geoscientific community. At the recent COP28, our data were featured to a global audience in a live demo session, attended by over 100 stakeholders and decision-makers in geothermal energy, highlighting our project's relevance in sustainable energy developments. Notably, our data contributed to a collaborative effort resulting in the mapping of economically feasible geothermal hotspots worldwide.

5.6 Evaluation of Project Success

Our project success evaluation is designed towards different criteria related to data and metadata and the new research data infrastructure itself.

Criteria 1: Data Quality and Management

To ensure that the reported heat-flow data are accompanied by rich metadata and information for quality assessment, we have established a **new community-driven protocol and metadata schema,** alongside measures for data quality (Fuchs et al., 2023). These innovations include **new data upload templates** designed to align with this schema (IHFC et al., 2024). Developed to comply with the FAIR principles, the schema incorporates persistent identifiers such as ORCID, ROR, IGSN, and DOIs for persons, institutions, samples, data and related literature. It also includes published controlled linked-data vocabularies (SKOS/RDF) to support metadata interoperability and properties that facilitate data quality assessment. Additionally, **codes** were developed and published enabling automatic vocabulary checks and quality scoring of data in the template format (Chishti et al., in prep). These measures collectively ensure that the data is not only of high quality but also easily accessible and usable by the geoscientific community.

Criteria 2: Technical Performance and user engagement

Since the release of the prototype is planned after the submission of this report (M28) for M35 of phase 1, only limited aspects of technical performance and user engagement can be evaluated at this moment. The stakeholder feedback on the technical performance of a release preview was positive (cf. section 5.5), engaging us to continuously follow the proposed path. The engagement of future key users in terms of the data revision is enormous and the essential backbone towards a future authenticated heat-flow database.

Criteria 3: Impact on Research

The project has supported the already existing international collaborations in terms of the standardization and the assessment and extension of research data and metadata for heat flow and geothermal applications. Institutes from more than 40 countries and 6 continents have been involved in the conceptual and the data-driven work since the project started. Guest research stays of several months from 13 researchers have been realized at 4 international collaborating institutes. The data publications of global heat-flow data released from this project increasingly build the base for many research studies on global thermal aspects – benefiting from the large collaborative network – and will form into the new standard compilation in the following years.

Criteria 4: Sustainability and Long-term Viability:

Already before the project started, the GFZ (signed by the CEO) committed to the sustainable and long-term operation of this data infrastructure in-house. This includes all required financial, personal, or organizational support required for the operation, maintenance, and continuous update of the infrastructure beyond the initial grant period. On the expert data level, our transparent collaborative community approach will pay off in the future, as it ensures a high level of interaction and utilization of our research data infrastructure and the derived data products as well as a community and stakeholders interested in continuous improvements.

Criteria 5: Knowledge Transfer and Capacity Building:

Our scientific conference (Cermak7; June 2022) and the training programs (summer school 2023, workshops, and online webinars) were attended by 120+ senior and young researchers. Our YouTube channel with introductory videos has 60 subscribers and, since the project started, we have welcomed an increasing number of guests researchers in our unit at GFZ.